

AI Assisted Upgrade of a Parallel Circuit Power Box Simulation

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Abstract

This project focuses on creating and analyzing an AI Physics Lab simulation called “Polarized Light - From Wave to Image”, made by Claude AI running on Sonnet 5, that demonstrates the polarization and filtering of electromagnetic waves. The simulation uses a 3D interactive model of a projector, polarizing filters, and a screen to show how changing the filter angles affects the final light intensity and appearance. By rotating the filters, users can observe how waves are filtered and understand the relationship between filter angles and light intensity using concepts such as Malus's Law. The relationship is described by the formula:

Abstract

This project focused on improving a parallel circuit simulation with help from artificial intelligence. The original simulation used as a reference was the Boston University “Power boxes: parallel circuit” simulation, which shows how power is distributed in a parallel circuit. My goal was to make the simulation more interactive and easier to understand for students.

The upgraded simulation includes branch switches, battery internal resistance, voltage labels, animated current flow, live equations, and power boxes that change size as the circuit changes. I also revised the circuit drawing so it clearly does not look like a short circuit. The final version helps users see how equivalent resistance, current, voltage, and power are related in a parallel circuit. This project also showed how AI can be useful for creating, testing, and revising educational physics simulations.

Introduction

The physics topic I chose for this project was parallel circuits. A parallel circuit is a circuit where there are multiple paths for current to flow. Each branch is connected across the same two battery terminals, so each closed branch has the same voltage across it. However, the current in each branch can be different depending on the resistance in that branch. The total current from the battery is the sum of the currents in all of the branches. This makes parallel circuits different from series circuits, where the same current flows through every component.

The reference simulation I used was the Boston University “Power boxes: parallel circuit” simulation by Andrew Duffy. That simulation shows a battery connected to three resistors in parallel and uses power boxes to represent how much power is being used by each resistor. It is a useful simulation because it connects the mathematical idea of power to a visual area. When the power is larger, the box is larger. This helps make the physics easier to see instead of only reading numbers or equations.

Even though the original simulation was useful, I wanted to improve it by making it more interactive. In the original version, the user could change values and see power boxes, but I wanted to add more features that would help students understand what is physically happening in the circuit. My upgraded version adds switches to each branch, so the user can open or close individual paths. It also adds internal resistance to the battery, which makes the simulation more realistic because real batteries are not perfect. Another important change was adding voltage labels, current labels, animated current flow, and live equations at the bottom of the simulation.

A major goal was to make sure the circuit was visually clear and not confusing. At one point, the circuit drawing looked like it might be a short circuit because some of the wires seemed to create a direct path around the resistors. A short circuit would mean current can flow through a path with almost no resistance, which would be unrealistic and unsafe in a real circuit. I revised the drawing so the positive node and negative return are separated. This makes it clear that current must pass through one of the resistor branches. Overall, the purpose of the upgraded simulation was to make parallel circuits easier to understand by combining visuals, controls, animations, and equations.

Methods

To develop the simulation, I used ChatGPT by OpenAI. The AI model used was GPT 5.5 Thinking. I used ChatGPT through the ChatGPT website in July 2026. ChatGPT helped write and revise the HTML, CSS, and JavaScript code for the simulation. The simulation was made as a single HTML file, which means it can run in a web browser without needing a separate program.

The first step was giving ChatGPT the link to the original Boston University simulation. I explained that I wanted to upgrade the simulation using JavaScript. My main request was to add individual branch switches, a slider for battery internal resistance, voltage labels, animations, and live equations. I also asked for the simulation to show power boxes that update in real time. ChatGPT generated the first version of the code with these features included.

After that, I tested the simulation by opening it in a browser. I moved the sliders, opened and closed the switches, and checked whether the numbers and visuals changed. This testing was important because the first version worked, but some parts of the circuit drawing were not clear enough. Some resistor connections looked slightly disconnected from the wires. This made the circuit look less realistic. I then asked ChatGPT to fix the resistor drawing so the resistors connected smoothly to the wires.

Another problem I noticed was that the circuit could look like a short circuit. Since a short circuit would not be a normal working parallel circuit, I asked ChatGPT to change the circuit so it was definitely not a short circuit. The revised version separated the positive side and negative return side of the circuit. This made the current path clearer because each branch now had to go through a resistor before returning to the battery. I also asked ChatGPT to make the green switches slightly longer to the left so they connected cleanly to the branch wires. This final change helped the circuit look more complete and professional.

The AI generated most of the code, but I still had to review the output and decide what needed to be fixed. I gave feedback after testing the simulation and asked for revisions when something looked wrong. This means the process was not just AI making the project by itself. It was a back and forth process where I tested the simulation, found issues, and used AI to help improve them.

Results

The final result is a working parallel circuit simulation made with HTML and JavaScript. The simulation shows a battery connected to three parallel resistor branches. Each branch has a resistor and a green switch. The switches can be opened or closed, which lets the user see how the circuit changes when one path is removed. When a branch is opened, current no longer flows through that branch, but the other closed branches can still work.

The simulation also has sliders that let the user change the battery voltage, the internal resistance of the battery, and the resistance values of the three branches. When the user changes a value, the circuit updates immediately. The current values, voltage values, equivalent resistance, and power values all change in real time. This makes the simulation useful because the user can directly connect the equation results to the circuit behavior.

One important feature is the internal resistance slider. In an ideal circuit, the battery voltage is usually

treated as constant. However, real batteries have internal resistance. When more current is drawn from the battery, some voltage is lost inside the battery. This causes the terminal voltage to drop. The simulation shows this by changing the terminal voltage label and shrinking the power boxes when internal resistance affects the circuit. This helps show why adding more parallel branches can change the total battery current and reduce the voltage available to the branches when internal resistance is included.

The power boxes are another main result of the project. The battery power box, internal resistance power box, load total power box, and individual branch power boxes all change size based on the calculated power. This makes the idea of electrical power easier to visualize.

Instead of just reading watts as a number, the user can see the box area grow or shrink. This is helpful for understanding that power depends on both voltage and current.

The final circuit drawing was also improved. The positive node and negative return are separated, and the current must pass through one of the resistor branches. The green switches were adjusted so they connect better to the wires. The resistors also have smoother connections to the branch wires. These drawing changes make the simulation look cleaner and make it easier to understand the actual circuit layout.

Discussion

The biggest strength of the simulation is that it connects the physics equations to a visual model. Many students can solve circuit equations but still have trouble picturing what is happening in the circuit. This simulation helps with that because the user can see the circuit, the switches, the current labels, the voltage labels, and the power boxes all changing together. This makes the simulation more interactive than a static diagram.

Another strength is that the simulation includes battery internal resistance. This is useful because it makes the circuit more realistic than a perfect ideal battery. When internal resistance is added, the terminal voltage can drop when the total current increases. This helps explain why real circuits do not always behave exactly like simple ideal circuit diagrams. The live equations at the bottom also help because they show how equivalent resistance, total current, terminal voltage, and power are being calculated.

The branch switches are also helpful. When a user opens one branch, they can see that the other branches still have a path for current. This is one of the key ideas of a parallel circuit. In a series circuit, opening one part would stop the whole circuit. In a parallel circuit, opening one branch does not necessarily stop current in the other branches. The simulation makes this easier to see.

One limitation is that the simulation is still simplified. It does not include every detail of a real circuit. For example, it does not show heat building up in wires or resistors. It also does not simulate a real battery failing or becoming damaged. The animated current dots are only a visual representation, not a real model of electron motion. Also, the simulation only has three branches, so it is limited compared to larger or more complex circuits.

In the future, I would improve the simulation by adding graphs. For example, a graph could show terminal voltage versus total current, or power versus resistance. I would also add a tutorial mode that explains each part of the simulation step by step. Another possible improvement would be adding meters, like a voltmeter and ammeter, that the user could drag around the circuit. This would make the simulation feel more like a real lab activity. I would also add warnings if the user creates an unsafe setup, like very low resistance with high current.

Overall, this project helped me understand both the physics and the coding process better. I learned that even if AI can create code quickly, the user still needs to test it carefully. The first version was not

perfect, and I had to notice problems with the circuit drawing and ask for changes. This showed that AI is useful, but human review is still important.

Conclusion

In this project, I used AI to upgrade a parallel circuit power box simulation. The final simulation includes branch switches, battery internal resistance, voltage labels, animated current flow, live equations, and changing power boxes. I also fixed the circuit drawing so it clearly does not look like a short circuit. The upgraded simulation helps show how resistance, current, voltage, and power are connected in a parallel circuit.

This project helped me learn more about parallel circuits and how simulations can make physics easier to understand. It also showed me how AI can be used as a tool for coding and improving a project. ChatGPT helped create the simulation, but I still had to test it, find problems, and ask for corrections. In the future, I would improve the simulation by adding graphs, more branches, moveable meters, and a better tutorial mode.

References

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